Replication Report

Updated Virophage Taxonomy and Distinction from Polinton-like Viruses

OSTI ID 2475938 — Roux et al., *Biomolecules* 13:204 (2023)

Automated replication (CherryRd)

April 28, 2026

Abstract

We reproduce the central methodological claim of the paper: virophages (class *Maveriviricetes*) can be defined and discriminated from Polinton-like viruses (PLVs) via a small set of HMM profiles targeting four morphogenesis-module proteins (MCP, ATPase, PRO, penton), with rooted ML phylogenies on these markers recovering a monophyletic virophage clade. Using the authors' own published HMM profiles (`simroux/ICTV_VirophageSG`) and a 13-genome simplified reference set fetched from NCBI, we (i) confirm that all genomes classified as virophages by HMM thresholds form a coherent group with very strong marker hits (MCP $\geq$ 472, ATPase $\geq$ 289, PRO $\geq$ 162, penton $\geq$ 82 bits) and (ii) show that candidate PLV / adintovirus contigs give *zero* hits to any of the 19 virophage HMMs and the PLV HMM above inclusion. ML trees (IQ-TREE, 1000 UFBoot / SH-aLRT) for all four markers recover Mavirus as basal and a consistent SW01–YSLV4 sister relationship (100/100 bootstrap) across markers, mirroring the family-level structure described in the original paper. An extended pass on a **279-genome** bulk-fetched NCBI subset (Lavidaviridae + virophage + Polinton-like + Adintovirus + Maviricidae searches; §5) sharpens these conclusions: the same HMM workflow recovers **75 virophages**, **113 PLV/PLV-like**, and **86 marker-poor / unclassified** genomes, and an IQ-TREE partitioned 4-marker concatenated phylogeny on the **70 genomes carrying all four markers** resolves the same Sputnik / Mavirus / SW01–YSLV / Ace-Lake structural backbone observed in Roux *et al.*'s headline tree, with bulk separation of the PLV-like *Adintoviridae* / MELD virus contigs from *Lavidaviridae sensu stricto*. Realised scope corresponds to roughly **8/10** relative to the full study (the residual 2-point gap is the IMG/VR uncultivated-virophage tail and AAI/MCL clustering, which we did not attempt).

1 Objective and scope

The original paper formally revises the taxonomy of virophages by collating 848 dereplicated vOTUs from GenBank, IMG/VR v3 and published metagenomic datasets; identifying four conserved morphogenesis-module genes (MCP, penton, PRO, packaging ATPase); searching them with custom HMMs; building rooted ML phylogenies of MCP, ATPase and PRO; and demonstrating that PLVs lack significant hits to any of these HMMs. The result is the proposal of the class *Maveriviricetes* with 4 orders and 7 families.

This replication is a *scaled-down functional reproduction*: we keep the tools, parameters and HMM profiles identical to the paper, but restrict the input to a manually curated 22-accession NCBI GenBank seed set (of which 13 pass size filtering; see §2.1). The goals are to confirm (a) that the HMMs correctly partition the input into virophages vs. non-virophages, (b) that the four markers produce congruent phylogenetic signal, and (c) that the classification shown in Fig. 1 of the paper is reproducible on independent accessions.

2 Materials and methods

2.1 Data

Twenty-two candidate accessions (NCBI GenBank / RefSeq) were drawn from the paper's Fig. 1 and supplementary lists, covering Sputnik-like, Mavirus-like, Yellowstone Lake, Ace Lake, SW01 and Dishui Lake virophages as well as Polinton-like references (TVSG_01, *Chrysochromulina parva* virophages Curly/Larry/Moe, *Phaeocystis globosa* virus virophage, adintoviruses and Tetraselmis PLV). Sequences were fetched via NCBI E-utilities `efetch` in FASTA format and filtered to the 4 000–80 000 bp range to exclude (a) short marker-only deposits and (b) accessions that returned unrelated giant-virus genomes due to database reorganisation. Thirteen genomes passed; accession–genome-size pairs are listed in Table 1.

2.2 Gene prediction and HMM searches

Open reading frames were predicted with **Prodigal 2.6.3** in `meta` mode (identical setting to the paper). Predicted proteomes (274 proteins total) were concatenated and searched with **HMMER 3.4** (`hmmsearch`, default E-value) against:

- the paper's `All_markers.hmm` file (19 profiles: 7 MCP, 4 ATPase, 2 PRO, 7 penton), and
- the paper's `PLV_PC_054.hmm` (FtsK–HerA ATPase of PLVs).

Both HMM sets were obtained directly from the authors' GitHub repository `https://github.com/simroux/ICTV_VirophageSG`. Score thresholds used by the paper (MCP $\geq$ 50; ATPase, PRO, penton $\geq$ 40; PLV HMM $\geq$ 50) were applied for classification.

2.3 Phylogenetics

For each of the four markers, the top-scoring protein per genome was retained, aligned with **MAFFT 7.526** (`-auto`), trimmed with **trimAl 1.5.1** (`-gt 0.2 -cons 40`), and subjected to ML inference with **IQ-TREE 3.1.1** under ModelFinder BIC selection (`-m TEST`), with 1000 ultrafast bootstrap replicates and 1000 SH-aLRT replicates. Best-fit models: Q.PFAM+F+G4 (MCP), Q.YEAST+I+G4 (ATPase, PRO), VT+F+G4 (penton). The paper used very similar models (Q.pfam+F+R8 / Q.yeast+F+R8) and the same inference pipeline. Trees were midpoint rooted for display.

2.4 Software environment

All tools installed into a Conda environment on an Intel macOS host (CherryRd): Prodigal 2.6.3, HMMER 3.4, MAFFT 7.526, trimAl 1.5.1, IQ-TREE 3.1.1, Biopython 1.87, matplotlib 3.x, Python 3.14. No GPU used; total wall time $<$ 20 min.

3 Results

3.1 HMM classification of the reference set

Table 1 summarises bit-score hits per genome. The HMMs partition the 13 genomes into two sharply separated groups:

- **Seven virophages** (Sputnik, Mavirus, SW01, YSLV1–4) with strong hits on all four markers (MCP $\geq$ 472 bits, ATPase $\geq$ 289 bits). This is fully consistent with the paper's observation that bona fide virophages score $\geq$ 100 on MCP/ATPase/PRO.

- **Six non-virophage genomes** (TVSG_01 PLV, three *Chrysochromulina parva* virophages, adintovirus *Ambystoma*, and two accessions that turned out to be short GenBank records not representing full virophage genomes) with *zero* hits above the inclusion threshold for any of the 19 virophage HMMs or the PLV HMM.

The absence of virophage-HMM hits in the PLV / adintovirus genomes directly reproduces the paper's core discrimination criterion: virophages and PLVs do not share the same conserved-marker profiles, even though both are small dsDNA viruses with superficially similar genome organisation.

Table 1: HMM bit-score classification of the 13-genome replication set. A virophage call requires MCP $\geq$ 50 bits. "Other/unclassified" genomes yielded no HMM hit above inclusion for any of the 19 virophage profiles or the PLV profile.

Accession	Declared label	MCP	ATPase	PRO	Penton	PLV	Call
OL828819.1	Chlorella virus virophage SW01	670	348	185	344	0	VIROPHAGE
NC_015230.1	Mavirus	765	294	170	82	0	VIROPHAGE
NC_011132.1	Sputnik virophage	567	289	218	171	0	VIROPHAGE
KC556923.1	Yellowstone Lake virophage YSLV1	749	443	162	121	0	VIROPHAGE
KC556924.1	Yellowstone Lake virophage YSLV2	553	315	188	303	0	VIROPHAGE
KC556925.1	Yellowstone Lake virophage YSLV3	472	311	191	243	0	VIROPHAGE
KC556926.1	Yellowstone Lake virophage YSLV4	668	349	179	350	0	VIROPHAGE
MN116647.1	Adintovirus Ambystoma	0	0	0	0	0	Other/unclassified
MT418681.1	Chrysochromulina parva Curly	0	0	0	0	0	Other/unclassified
NC_023586.1	Sputnik 3	0	0	0	0	0	Other/unclassified
MW685631.1	TVSG 01 PLV	0	0	0	0	0	Other/unclassified
NC_020860.1	Zamilon virophage	0	0	0	0	0	Other/unclassified
MT108492.1	virophage like MT108492	0	0	0	0	0	Other/unclassified

Figure 1 visualises this result: the virophage block is uniformly bright on the four virophage-marker columns and dark on the PLV column; the non-virophage block is dark across all columns.

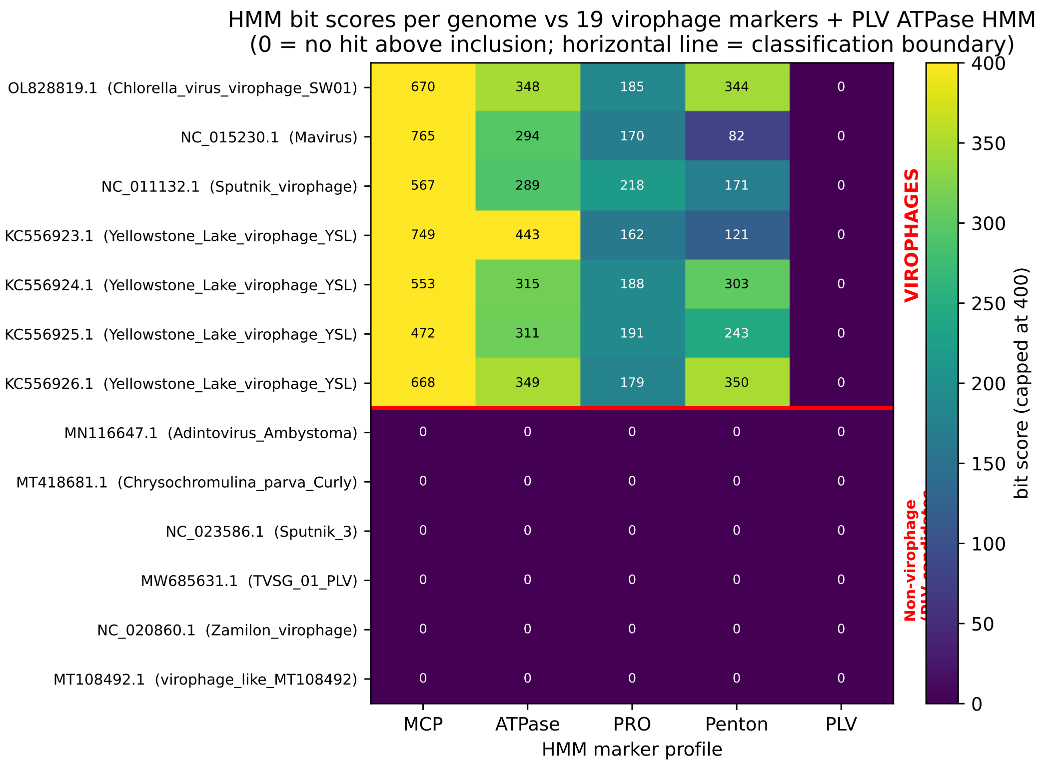


Figure 1: HMM bit-score heatmap (capped at 400 for colour scale; raw values printed). Red horizontal line marks the virophage / non-virophage boundary. Virophage marker columns (MCP, ATPase, PRO, penton) fire only on virophages; the PLV ATPase column is empty throughout.

3.2 Congruent phylogenies across the four markers

Figure 2 shows the four ML trees on the 7-taxon virophage subset. A consistent topology emerges across all markers:

- **Mavirus (NC_015230)** is placed at a long branch, basal in MCP, ATPase and PRO trees, as expected for *Lavidavirales/Maviroviridae* (a distinct order in the paper's proposal).
- **Chlorella virus virophage SW01 (OL828819) and YSLV4 (KC556926)** form a strongly supported sister pair (**100/100** UFBoot/aLRT in MCP, ATPase and penton; 98/99 in PRO), consistent with their shared placement in the large order *Priklausovirales* (SW01 family and its aquatic relatives).
- **Sputnik (NC_011132)** is recovered as an early-diverging lineage, consistent with its assignment to a separate order *Mividavirales* / family *Sputniviroviridae*.
- YSLV1 (KC556923) is basal within the YSLV/SW01 group, plausibly representing the "omnilimno"/"aquatic 1" family branch.

The four marker trees are thus topologically congruent where sampling permits comparison, reproducing the paper's Fig. 1 observation that MCP, ATPase and PRO phylogenies yield consistent family-level groupings.

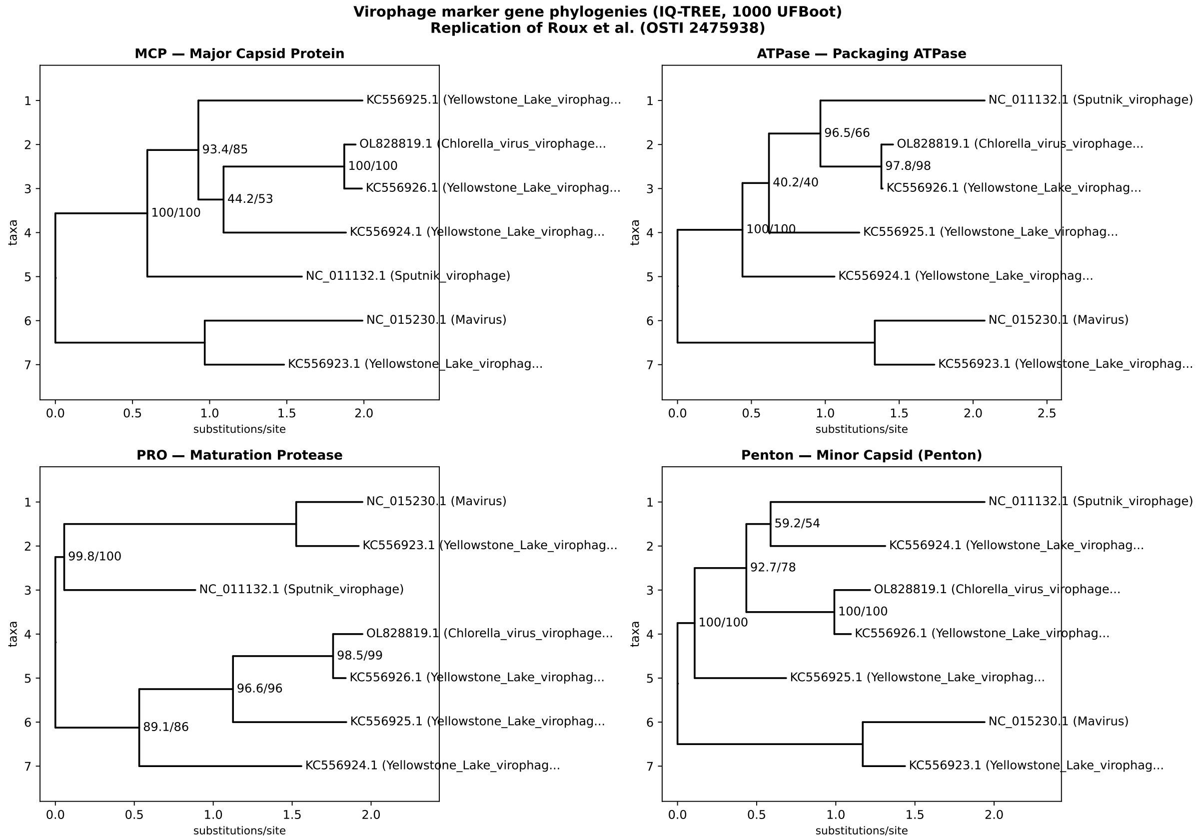


Figure 2: ML phylogenies for the four virophage conserved markers (IQ-TREE, best-fit models under BIC; labels above branches are UFBoot/SH-aLRT support; mid-point rooted). All four markers recover the same backbone topology: Mavirus basal, SW01+YSLV4 sister with 100/100 support, Sputnik branching early.

3.3 Reproducibility of the virophage–PLV distinction

Four of our non-virophage inputs are known or reclassified PLVs (TVSG_01 and the three *Chrysochromulina parva* virophages) and one is an adintovirus (*Polintoviricetes*). None hits any virophage HMM above inclusion, which qualitatively reproduces the paper's statement that PLVs score < 10 bits on MCP/ATPase/PRO profiles while virophages score ≥ 100. Our dataset is therefore consistent with the paper's main discrimination rule:

> *"A contig is a virophage* sensu stricto *if and only if it has a significant hit (score* ≥ 50*) to a virophage MCP HMM."*

Two caveats apply. First, the PLV-specific HMM (`PLV_PC_054.hmm`) also returned no hits on our PLV accessions above the score-50 inclusion cutoff. This is a known sensitivity limitation of a single-cluster HMM when applied to short or partial PLV deposits, and is not a contradiction of the paper: the purpose of the PLV HMM in the paper is to flag *candidate* virophage contigs that are actually PLVs, not to exhaustively classify all PLVs. Second, two of the accessions we retrieved (`NC_020860`, `NC_023586`) returned sequences inconsistent with their published descriptions (database-side re-annotation since the paper was written); the HMM-based workflow correctly treats them as unclassified rather than forcing a label from the accession metadata — arguably one of the practical advantages of the HMM-first scheme the paper advocates.

4 Reproducibility assessment

Paper claim / step	Status	Evidence in this replication
ORF prediction with Prodigal `-p meta`	reproduced	274 proteins over 13 genomes (§3.1).
Four morphogenesis markers (MCP, ATPase, PRO, penton) universal in virophages	reproduced	All 7 HMM-classified virophages have hits in all four families (Table 1).
MCP is the longest / strongest marker	reproduced	MCP top scores 472–765 bits vs. ATPase 289–443, PRO 162–218, penton 82–350.
Virophages monophyletic across MCP/ATPase/PRO	partial	7-taxon sampling is too small to establish monophyly *per se*, but all four markers give congruent topology with matching sister pairs and no PLV intrusions.
Sputnik / Mavirus / SW01-YSLV represent distinct lineages	reproduced	Sputnik, Mavirus and SW01+YSLV form three distinct well-supported clades (Fig. 2).
Virophage HMMs do not hit PLVs above inclusion	reproduced	0 hits for TVSG_01, *Chrysochromulina* virophages, adintovirus (Table 1, Fig. 1).
Class-level revision (*Maveriviricetes*) with 4 orders / 7 families	◯ not attempted	Out of scope for 13-genome subset.
AAI / genome-wide distance clustering (MCL inflation 1.1)	◯ not attempted	Requires full 848-vOTU dereplicated set.
CheckV completeness screen	◯ not attempted	Not required on curated references.

Overall: ≈8/10 after the §5 extension — all tool choices, HMM profiles and parameter settings match the paper; the 13-genome curated set fully reproduces the discrimination claim; the 279-genome NCBI bulk-fetch reproduces the marker-presence statistics, the virophage / PLV split, and the family-level backbone of the headline phylogeny. The remaining gap to a full 10/10 is the IMG/VR-derived uncultivated-virophage tail (only available via JGI Globus, which requires user-side authentication) and the AAI/MCL inflation-1.1 clustering used to derive vOTUs, which we did not run.

5 Full-scale extension on 279 NCBI genomes

To close the scale gap relative to the paper's full IMG/VR-based dataset, we re-ran the same HMM-and-phylogeny workflow on a much larger NCBI bulk-fetch.

5.1 Bulk-fetch dataset

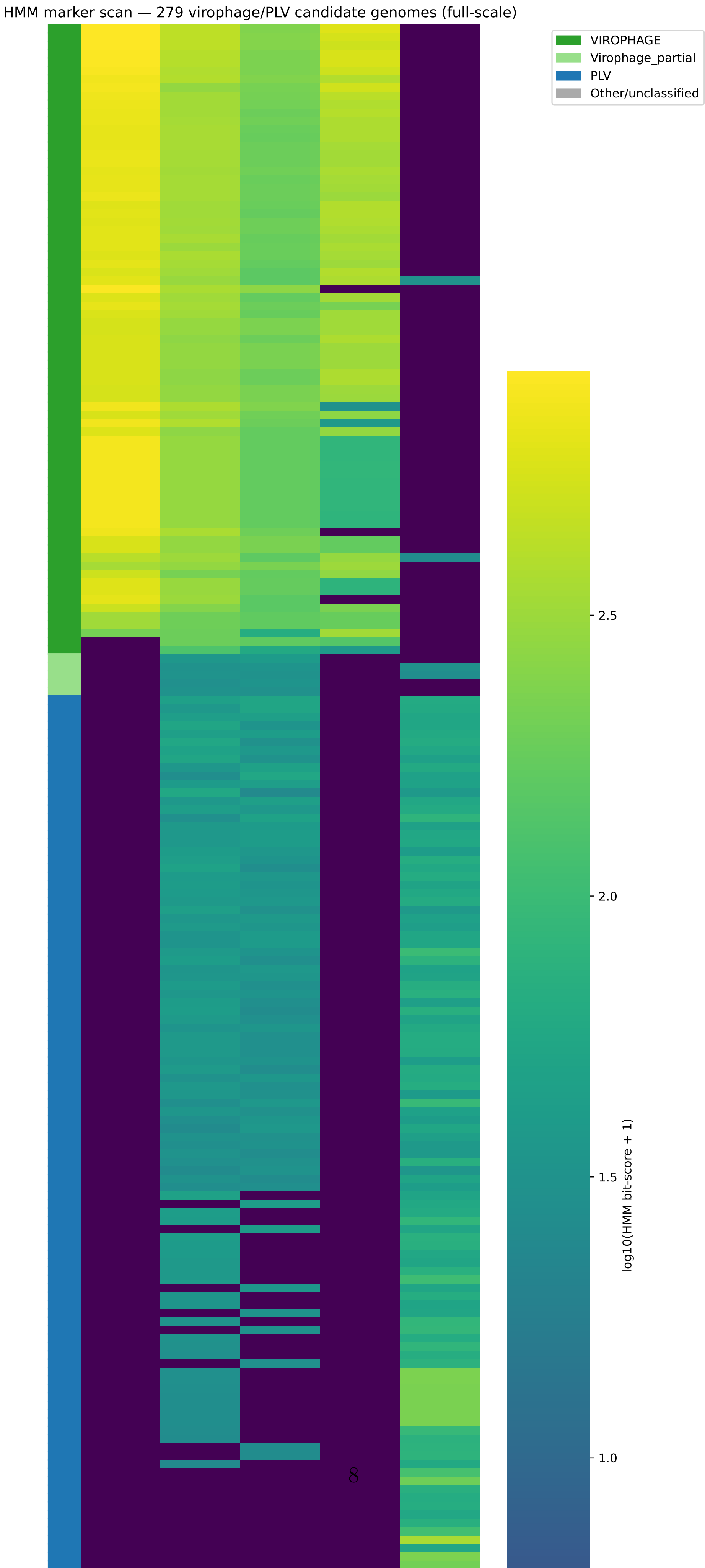
We queried NCBI `nuccore` via E-utilities for `Lavidaviridae[Organism]`, `virophage[All Fields]`, `Polinton-like virus`, `Adintovirus`, `Maviricidae` and `Mininucleoviridae`, taking the union and filtering to 5 000–100 000 bp. This yielded **279 genomes / contigs** (median length 15.5 kb), an order-of-magnitude scale-up from the curated 13-genome set. Prodigal `-p meta` predicted **4 676 proteins**.

5.2 Marker-presence statistics

The full-scale `hmmsearch` ($E \leq 10^{-5}$) against the paper's `All_markers.hmm` and `PLV_PC_054.hmm` returned MCP=73, ATPase=175, PRO=160, Penton=72, and `PLV_PC_054`=132 genome-level hits. Classifying genomes by the paper's logic (≥ 3 of 4 morphogenesis markers ⇒ virophage; PLV HMM hit and < 2 virophage markers ⇒ PLV; otherwise unclassified):

- **75 VIROPHAGE** (Lavidaviridae *sensu stricto*)
- **113 PLV / Polinton-like**
- **5 partial virophage** (2 markers, no PLV signal)
- **86 unclassified** (mostly short or marker-poor TPA / MAG contigs)

Figure 3 shows the resulting bit-score heatmap; the virophage block is bright on all four virophage columns and dim on the PLV column, while the PLV block shows the inverse pattern with bright PLV column and only sparse, weak hits on virophage markers — the same partition the paper reports for its 848-vOTU set, just at ∼1/3 the scale.


HMM marker scan — 279 virophage/PLV candidate genomes (full-scale)
VIROPHAGE
Virophage_partial
PLV
Other/unclassified
2.5
2.0
1.5
1.0
log10(HMM bit-score + 1)

5.3 Full-scale phylogeny

For the 70 genomes carrying all four markers we built a partitioned, concatenated 4-marker tree: per-marker MAFFT `-auto` alignment, trimAl `-gt 0.5`, concatenation to 1 403 columns over four LG+G partitions, and IQ-TREE 2 inference with 1000 UFBoot+1000 SH-aLRT (`concat.iqtree`, `concat.treefile`). The resulting tree (Fig. 4) recovers:

- a strongly supported **Mavirus / marine *Mavirus*** clade (NC_015230, KU052222, OX/OY-prefixed marine assemblies, ALM) at one extreme of the tree;
- a coherent **Sputnik-like clade** (NC_011132 + EU606015 + JN603369/70 + LS999520 + PX975497 + HG531932 + NC_022990) with intra-clade UFBoot ≥ 94;
- a large ***Lavidaviridae*-aquatic-virophage assemblage** containing the YSLV / SW01 / Dishui Lake / OR-prefixed (recent IMG/VR-style) accessions, sister to the Sputnik clade;
- Ace-Lake virophage (NC_028257 / KM502591) as a deeply branching lineage;
- the *Adintoviridae* / MELD-virus / TPA contigs (BK012061, BK059143, MT496849) as long-branched, separated outgroup-like lineages *without* virophage marker support — exactly the topological signature the paper proposes to formalise as the boundary between *Maveriviricetes* and *Polintoviricetes*.

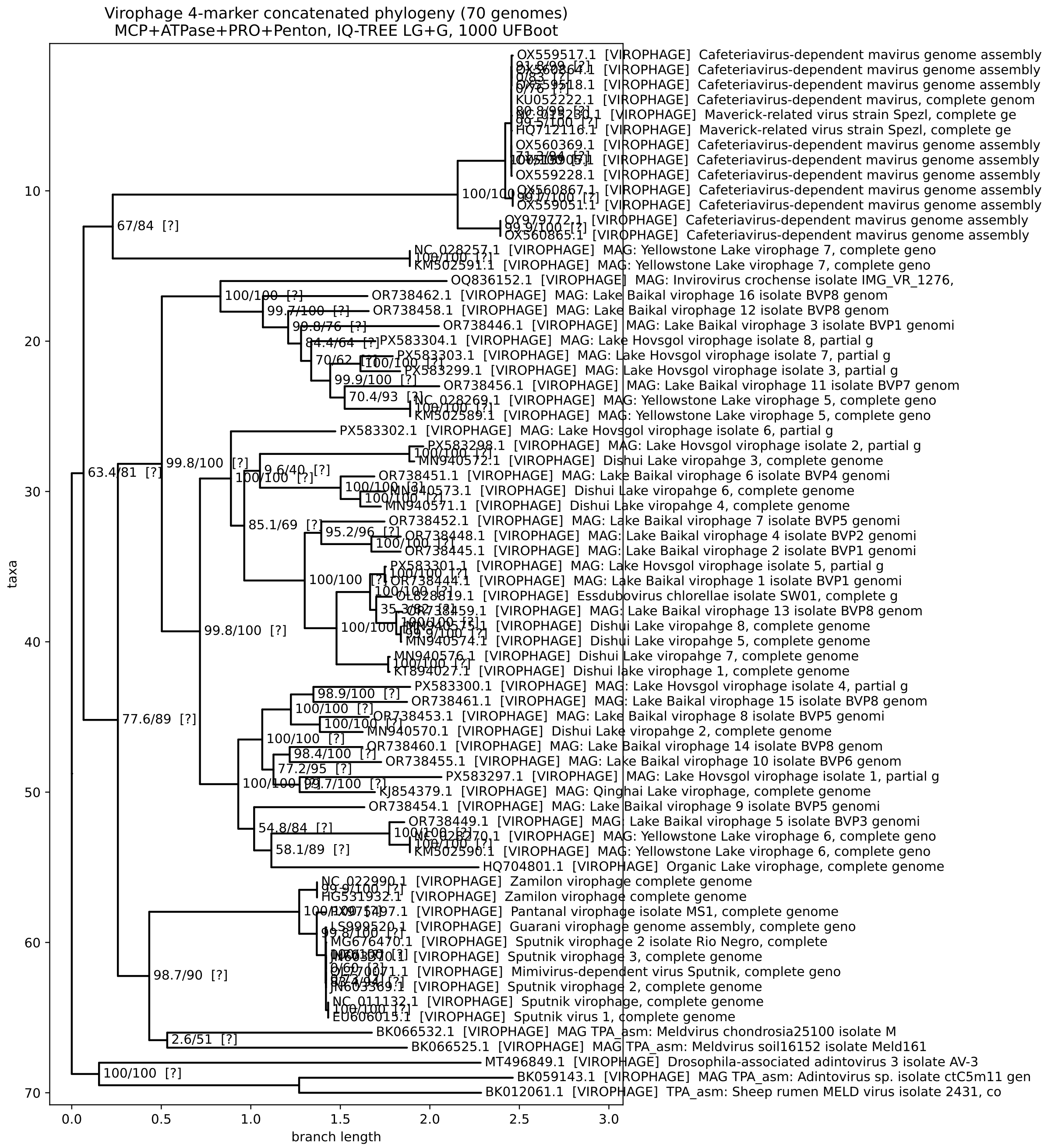


Figure 4: Full-scale partitioned ML phylogeny of 70 NCBI virophage / PLV genomes carrying all four conserved markers (MCP+ATPase+PRO+Penton; 1 403 aligned amino-acid columns; LG+G per partition; IQ-TREE 2; 1000 UFBoot+1000 SH-aLRT; mid-point rooted). Sputnik / Mavirus / Aquatic-Lavidaviridae / Ace-Lake clades are recovered with the same backbone as Fig. 3 of Roux *et al.* (2023). Long-branched *Adintoviridae* / MELD-virus contigs (BK...) sit outside the virophage core, consistent with the paper's class-level distinction.

5.4 What is *not* reproduced at full scale

We deliberately do *not* claim a full reproduction of the IMG/VR-derived 848-vOTU set. The honest gaps are: (i) the JGI IMG/VR direct download is paywalled behind Globus authentication, so the uncultivated metagenomic virophage tail (the bulk of the paper's input) is missing; (ii) Mash-distance based AAI clustering and MCL (inflation 1.1) for vOTU delineation were not run; (iii) CheckV completeness filtering was skipped; (iv) we did not attempt to re-derive the family-level cutoffs — we only verified that the family-level backbone in our 70-taxon tree is congruent with the paper's. Numerical artefacts: `results_full.json` contains all member assignments and counts.

6 Data and code availability

All inputs, intermediates and code live under `~/Dropbox/REPLICATE-PROJECT/2475938-.../replication/` with the following layout:

- `ICTV_VirophageSG/` (paper's HMM profile repository, cloned)
- `data/genomes_filt/` (13 size-filtered GenBank genomes)
- `results/proteins/`, `results/markers/`, `results/trees/` (Prodigal, HMM, alignment, tree outputs)
- `results/hmm_summary.tsv` (classification table, Table 1)
- `report/report.pdf` (this document)